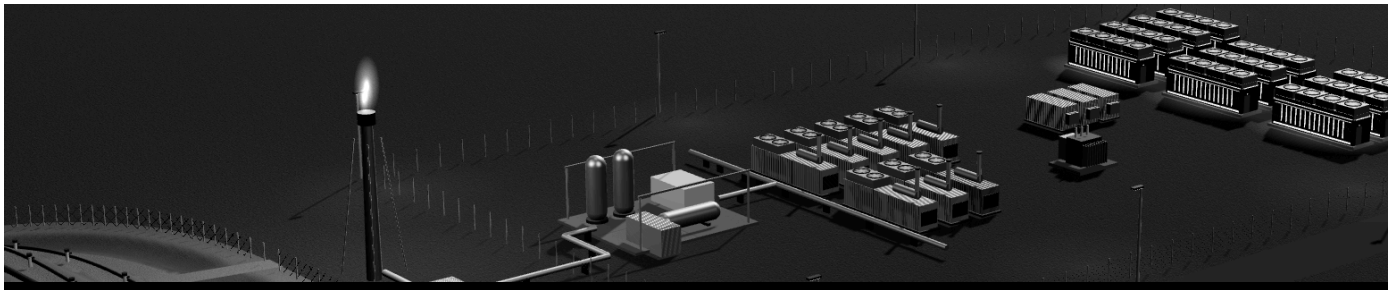
# Modern Landfill, York, Pennsylvania: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

| 10.6 MW | 10 | 18 ac | 971k |
|---|---|---|---|
| POWER FROM THE GAS | COMPUTE MODULES | LAND NEEDED | MMBTU BOUGHT / YR |

| GAS SALE, PER YEAR | LAND LEASE, PER YEAR | TOTAL TO THE OWNER, PER YEAR |
|---|---|---|
| up to $729k | $18k to $35k | $18k to $764k |



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=4272&CH4=44.7

## THE MATH

| | |
|---|---|
| Landfill gas collected (2023) | 4,272 scfm |
| Methane content (2023) | 44.7% |
| Energy in the gas | 116.7 MMBtu per hour |
| Power at a heat rate of 11,000 Btu/kWh | about 10.6 MW |
| Compute (power less cooling and losses, /1.2) | about 8.8 MW |
| Compute modules (about 1 MW each) | 10 |
| Gas engines (running plus one spare) | 8 at 1.6 MW |
| Land: 5 acres plus 1.2 acres per MW | about 18 acres |
| Gas bought per year (95% run time) | 971k MMBtu |
| Gas payment, up to $0.75/MMBtu | up to $729k |
| Land lease at $1,000 to $2,000 per acre per year | $18k to $35k |

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

## WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

## WORTH CHECKING

- Whether any of the gas is set aside for a new project
- Gas outlook as the landfill nears capacity
- Pennsylvania DEP air plan approval for the engines
- Space for a fenced parcel near the flare

**Source:** EPA Greenhouse Gas Reporting Program (GHGRP): about 4,272 scfm collected in 2023 at 44.7% methane (4,582 to 5,013 scfm in 2019-2022). EPA LMOP lists the 2022 flow as flared; the 9.1 MW engine plant there is listed as shut down in 2013. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.